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Hamza Mohammed Hassanain

Software Engineer

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EDUCATION

B.Sc. Computer and Information Sciences, Suez Canal University
AWS Certified Solutions Architect – Associate, Amazon Web Services

2022 – 2026
[credly.com/verify](https://www.credly.com/verify)

EXPERIENCE

Founding Software Engineer Nov 2025 – Present
Repovive (YC F’26) Full Time – United States, Remote
The science expert community for unsolved problems, starting with theoretical CS and math proofs; 10,000+ users.

- Joined as the **first engineer** and built the core systems from the early days through YC.
- Led design of the distributed judge (judge1 on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Built AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents run from versioned, admin-authored definitions.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build), no prod secrets.
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped the fixes.
- Built a real-time collaborative dev space using **WebSockets** and **Yjs CRDTs** over a Monaco live-syncing editor.

OPEN SOURCE CONTRIBUTIONS

- **GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | **!2692, !2691, !2675, !2670**.
- **GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | **!2881, !2909, !2852**.
- **GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | **!221163, !221158**.
- **LLVM / Clang**: `constexpr X86 cvtpd2ps`, `AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG` tests | **#169980, #178885, #178430, #173473**.
- **Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | **#3410**.

PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project github.com/sigma-loop
• Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
• Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge — AWS Reference Architecture github.com/HamzaHassanain/CP-Judge-AWS
• Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
• Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman
• Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress HALlocator | cppress
• Built a Red-Black-Tree best-fit memory allocator ($O(\log n)$) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Authored “What Every Programmer Should Know About Memory”, a series on CPU caches, virtual memory, and optimization.
- Published “I Built a Backend Web Framework from Scratch in C++”, covering non-blocking I/O, HTTP parsing, and thread pools.

ACHIEVEMENTS

ICPC ECPC & ACPC — Competitive Programming icpc.global/YC7Z02WQCPIB
17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.

SKILLS

Cloud & DevOps	AWS, Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux
Backend & Full-Stack	TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs)
Languages & Systems	C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues